Evan Hobson

evanjhobson@gmail.com | github.com/broncomcbuck | evanhobson.com

Education

Brigham Young University, Provo, UT — 2021 – Apr 2027 (expected)
B.S. Wildlife & Wildlands Conservation

Research Interests

Statistical and spatial modeling of rangeland and wildlife systems; habitat selection (RSF, SSF, iSSF); remote sensing of rangeland vegetation; grazing management on working landscapes.

Research Experience

BYU Rangeland Ecology Lab — **Provo, UT**
Research Assistant — Sep 2025 – Present

- Lead analyst for a GPS-collar study of cattle resource selection during dormant-season targeted grazing on a 998-ha cheatgrass-invaded sagebrush steppe allotment in southeastern Idaho (64 cow-years, 2021–2022, 5- to 7-minute fixes)
- Tested whether inferred cattle response to cheatgrass depends on the remote-sensing product by fitting identically structured RSFs that differ only in the annual-grass layer (RAP 10-m invasive annual grass, RAP 30-m annual forb and grass, USGS REPMS *Bromus tectorum*)
- Fit binomial GLMMs in glmmTMB with per-animal random slopes and year interactions on topography, distance to water and perennial forage; validated with animal-grouped cross-validation and the Boyce index
- Built a reproducible R pipeline from raw telemetry to validated models and selection maps
- Collected vegetation data for the Sagebrush Steppe Treatment Evaluation Project (SageSTEP) and a targeted sheep-grazing study (summer 2026): line-point intercept (BLM AIM), canopy and basal gap, density, biomass, shrub volume and soil stability

BYU Wildlife Ecology Lab — **Provo, UT**
Research Assistant — Jul 2024 – Jul 2025

- Pygmy rabbit camera trapping and habitat sampling; mule deer collar retrieval and cementum aging

Manuscripts

- **Hobson, E.**, et al. Cattle resource selection in cheatgrass-invaded rangelands during dormant-season grazing. In internal review with coauthors; target journal *Rangeland Ecology & Management*.

Awards

BYU College Undergraduate Research Award (CURA), $4,000, for the cattle resource-selection study

Professional Experience

Western North American Naturalist — **Provo, UT**
Assistant to the Editor-in-Chief — Apr 2025 – Jan 2026

- Coordinated more than 50 peer reviewers and tracked manuscripts through review and editorial decisions
- Managed disbursement of more than $10,000 in grant funds for journal operations

BYU Mathematics IT Department — **Provo, UT**
Lead Student Computer Specialist; faculty and lab support, Linux administration — May 2023 – Dec 2023

Other Writing

Hobson, E. Permission to feel, power to become: how emotional engagement can enable spiritual growth. BYU Student Religion Symposium; accepted for presentation and invited to the proceedings (not peer reviewed).

Skills

Statistics: GLMs and GLMMs (glmmTMB), conditional logistic regression, use–availability designs (RSF, SSF, iSSF), log-RSS, cluster-robust inference, cross-validation, Boyce index, spatial prediction
Computing: R (primary), Python, Bash, Git; ArcGIS; raster and vector workflows in R
Field: GPS and accelerometer collars, camera trapping, BLM AIM vegetation monitoring, plant functional-group identification